

Deep Learning Lab 2 Report

Movie Rating Prediction and Genre Analysis using NLP

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1 Objective

The objective of this lab is to explore Natural Language Processing techniques for movie metadata analysis. The tasks performed include:

- Text preprocessing of movie descriptions
- Word embedding using pre-trained GloVe vectors
- Movie rating prediction using neural networks
- Genre classification
- Genre-specific word analysis

The implementation uses PyTorch along with standard NLP libraries.

2 Dataset

The dataset contains movie metadata including:

- Movie overview
- Tagline
- Keywords
- Genres
- Average user rating

The dataset contains a total of:

4803 samples

Missing values were detected primarily in the `tagline`, `keywords`, and `genres` fields.

3 Dataset Split

The dataset was divided into training, validation, and test sets.

Dataset Split	Samples	Percentage
Training Set	3362	70%
Validation Set	720	15%
Test Set	721	15%

4 Text Preprocessing

The text fields used in the model include:

- Movie overview
- Tagline
- Keywords

Preprocessing steps:

- Lowercasing
- Tokenization
- Stopword removal
- Lemmatization
- Removing punctuation

These steps reduce noise and improve feature representation.

5 Word Embeddings

Pre-trained GloVe embeddings were used for representing words as dense vectors.

Metric	Value
Embedding Dimension	100
Total GloVe Vectors	400000
Dataset Vocabulary	17148
Words Covered by GloVe	15324
Embedding Coverage	89.36%

6 Feature Representation

TF-IDF vectorization was used to convert textual features into numerical representations.

The resulting TF-IDF matrix size was:

$$(3362, 5569)$$

which corresponds to the training dataset size and vocabulary size.

7 Baseline Model

A baseline model was implemented using the mean movie rating.

Metric	Value
Mean Rating	6.09
MSE	1.2825
RMSE	1.13248

8 Neural Network Models

Two neural models were trained to predict movie ratings based on text:

- Overview-based model
- Tagline-based model

The models use GloVe embeddings as input features.

Training was performed using mini-batch gradient descent with validation monitoring.

9 Model Comparison

The performance of different models was compared using RMSE.

Model	RMSE
Baseline Model	1.13248
Overview Model	1.21211
Tagline Model	1.16876

The baseline model performed slightly better than the neural models in this experiment.

10 Genre Classification

The dataset contains a total of **22 genres**, including:

- Action
- Adventure
- Animation
- Comedy
- Crime
- Documentary
- Drama

- Family
- Fantasy
- Fiction
- Foreign
- History
- Horror
- Movie
- Music
- Mystery
- Romance
- Science
- TV
- Thriller
- War
- Western

A multi-label classification model was trained to predict genres based on movie text descriptions.

11 Genre Word Analysis

The lab also analyzed words that are strongly associated with specific genres.

Examples include:

Genre	Rank	Indicative Word
Action	1	agent
Action	2	cop
Action	3	ruthless
Action	4	criminal
Action	5	assassin
Western	6	sheriff
Western	7	lawman
Western	8	bounty
Western	9	bandit
Western	10	notorious

This analysis helps identify words that strongly characterize different movie genres.

12 Conclusion

This lab explored several NLP techniques for analyzing movie metadata.

Key observations:

- GloVe embeddings successfully captured semantic information from movie descriptions.
- TF-IDF features provided an effective representation for text classification.
- Simple baseline methods can sometimes outperform neural models when the dataset is limited.
- Genre-indicative word analysis helps interpret model behavior.

Overall, this lab demonstrated how deep learning and NLP methods can be applied to movie metadata analysis and prediction tasks.